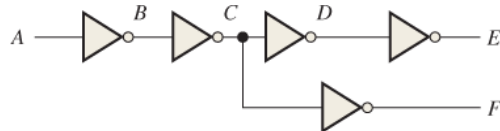


Chapter 3: 2, 5, 6, 11, 17, 22, 26.

2. A combination of inverters is shown in Figure 3–77. If a HIGH is applied to point *A*, determine the logic levels at points *B* through *F*.



$B = \overline{A}$, Since *A* is high *B* will be low

$C = \overline{B}$, Since *B* is low *C* will be high

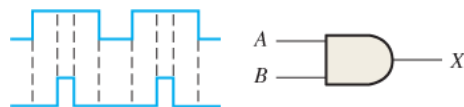
$D = \overline{C}$, Since *C* is high *D* will be low

$F = \overline{C}$ Since *C* is high *F* will be low

$E = \overline{D}$ Since *D* is low *E* will be high

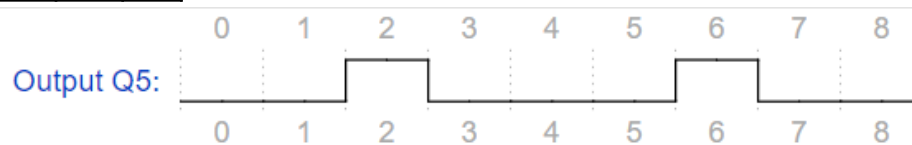
5. Determine the output, *X*, for a 2-input AND gate with the input waveforms shown in Figure 3–78.

Show the proper relationship of output to inputs with a timing diagram.

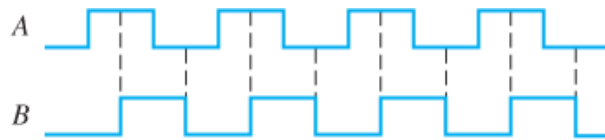


Truth Table:

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

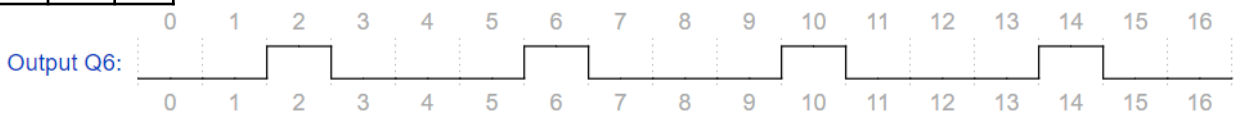


6. Repeat Problem 5 for the waveforms in Figure 3–79.

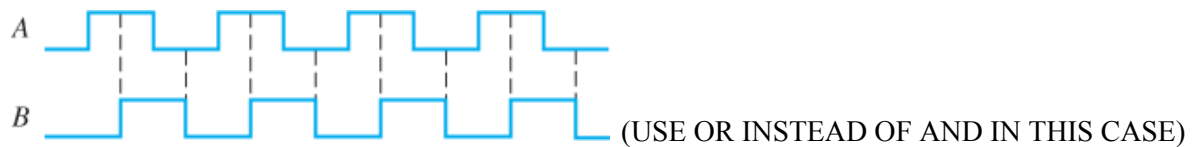


Truth Table:

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

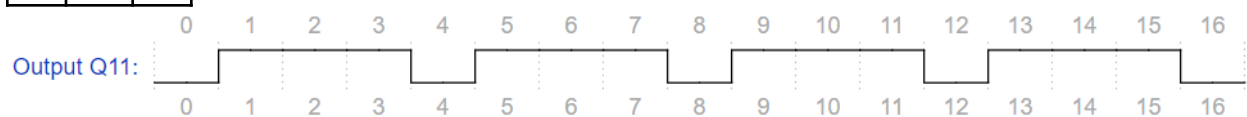


11. Determine the output for a 2-input OR gate when the input waveforms are as in Figure 3–79 and draw a timing diagram.

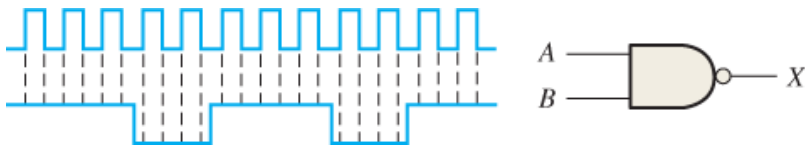


Truth Table:

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

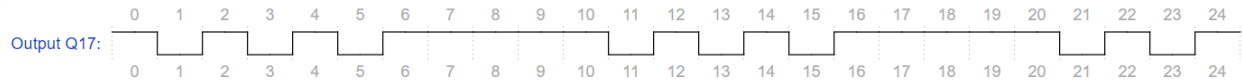


17. For the set of input waveforms in Figure 3–83, determine the output for the gate shown and draw the timing diagram.

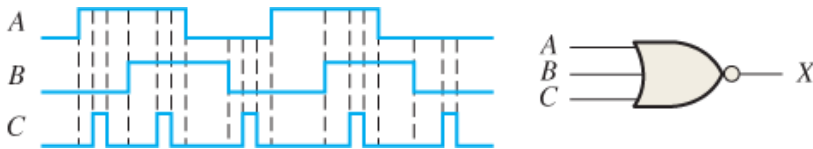


Truth table NAND

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

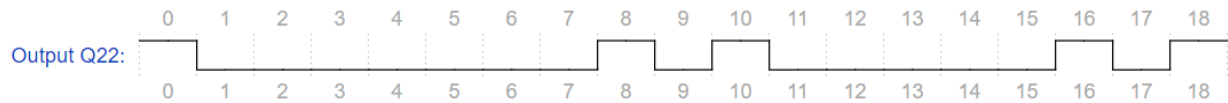


22. Determine the output waveform in Figure 3–87 and draw the timing diagram.

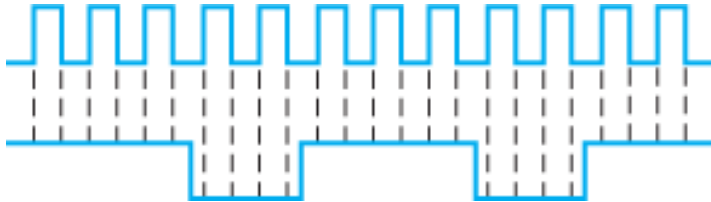


Truth table NOR

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

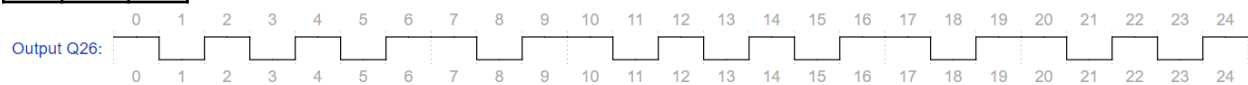


26. Repeat Problem 17 for an exclusive-OR gate.



Truth table XOR

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0



Chapter 4: 10, 15. Create your gate diagrams with waveDROM wherever possible.

10. Apply DeMorgan's theorems to each expression:

a. $\overline{A\bar{B}(C+\bar{D})}$

$$\begin{aligned}
 & \overline{A\bar{B}(C+\bar{D})} \\
 &= \overline{A\bar{B}} + \overline{C+\bar{D}} \\
 &= \overline{A} + \overline{\bar{B}} + \overline{C} + \overline{\bar{D}} \\
 &= \overline{A} + B + \overline{C} + D
 \end{aligned}$$

b. $\overline{AB(CD + EF)}$

$$\begin{aligned}
 \text{b) } & \overline{AB(CD + EF)} \\
 &= \overline{AB} + \overline{(CD + EF)} \\
 &= \overline{A} + \overline{B} + (\overline{CD} \cdot \overline{EF}) \\
 &= \boxed{(\overline{A} + \overline{B}) + ((\overline{C} + \overline{D}) \cdot (\overline{E} + \overline{F}))}
 \end{aligned}$$

c. $\overline{(A + \overline{B} + C + \overline{D})} + \overline{ABCD}$

$$\begin{aligned}
 \text{c) } & \overline{(A + \overline{B} + C + \overline{D})} + \overline{ABCD} \\
 &= \overline{A} \overline{B} \overline{C} \overline{D} + \overline{A} \overline{B} \overline{C} \overline{D} \\
 &= \boxed{\overline{A} + \overline{B} + \overline{C} + \overline{D}}
 \end{aligned}$$

d. $\overline{(\overline{A} + B + C + D)}(\overline{ABCD})$

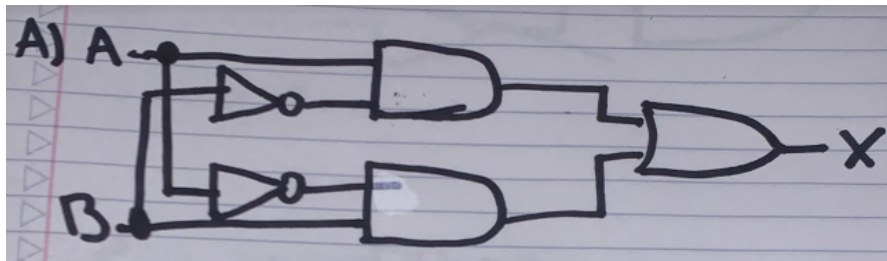
$$\begin{aligned}
 \text{d) } & \overline{(\overline{A} + B + C + D)}(\overline{ABCD}) \\
 &= \overline{\overline{A} + B + C + D} + \overline{ABCD} \\
 &= \overline{A} + \overline{B} + \overline{C} + \overline{D} + \overline{ABCD} \\
 &= \boxed{\overline{A} + \overline{B} + \overline{C} + \overline{D}}
 \end{aligned}$$

e. $\overline{AB}(CD + \overline{EF})(\overline{AB} + \overline{CD})$

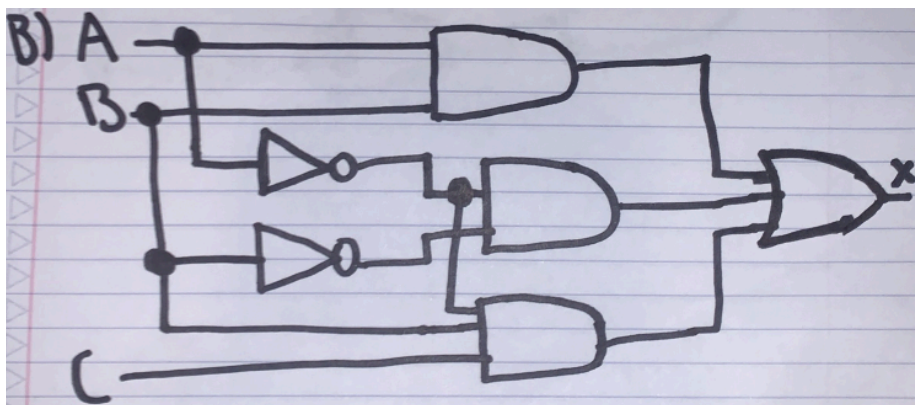
c) $\overline{AB}(CD + \overline{EF})(\overline{AB} + \overline{CD})$
 $= \overline{AB}(CD + \overline{EF}) + \overline{AB} + \overline{CD}$
 $= \overline{AB} + \overline{CD} + \overline{EF} + ABCD$
 $= \overline{AB} + \overline{CD} + \overline{EF} + ABCD$
 $= \overline{AB} + \overline{CD}(E + \overline{F})$
 $= \overline{AB} + \overline{CD}E + \overline{CD}\overline{F}$

15. Draw the logic circuit represented by each expression:

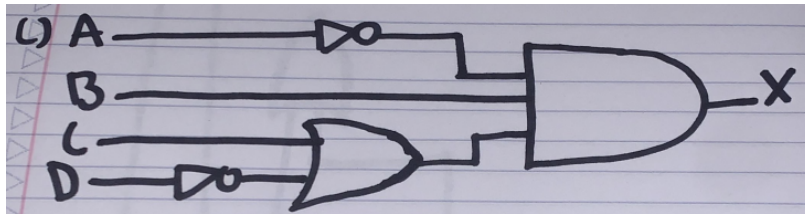
a. $\overline{AB} + \overline{A}B$



b. $AB + \overline{A}\overline{B} + \overline{A}BC$



c. $\overline{A}B(C + \overline{D})$



d. $A + B[C + D(B + \overline{C})]$

